

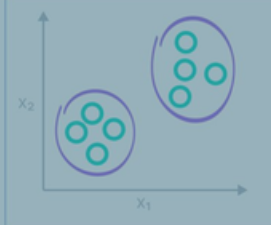


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**Unsupervised
Learning
(Clustering)**

Unsupervised Learning

Unsupervised learning is the training of a machine using information that is neither classified nor labeled and allowing the algorithm to act on that information without guidance. Here the task of the machine is to group unsorted information according to similarities, patterns, and differences.

Supervised learning	Unsupervised learning
Input data is labeled	Input data is unlabeled
Has a feedback mechanism	Has no feedback mechanism
Data is classified based on the training dataset	Assigns properties of given data to classify it
Divided into Regression & Classification	Divided into Clustering & Association
Used for prediction	Used for analysis
Algorithms include: decision trees, logistic regressions, support vector machine	Algorithms include: k-means clustering, hierarchical clustering, apriori algorithm
A known number of classes	A unknown number of classes
	

Unlike supervised learning, no teacher is provided, which means no training will be given to the machine. Therefore the machine is restricted to find the hidden structure in unlabeled data by itself.

Types of Unsupervised Learning

- Clustering is the type of Unsupervised Learning where we find hidden patterns in the data based on their similarities or differences.
- Association is the kind of Unsupervised Learning where we can find the relationship of one data item to another data item or discover the rules, such as people that buy X also tend to buy Y

Clustering

Clustering is the task of dividing the population or data points into a number of groups such that data points are more similar in the same groups and dissimilar in the other groups. It is basically a collection of objects on the basis of similarity and dissimilarity between them.

Applications of Clustering

- **Marketing:** It can be used to characterize & discover customer segments for marketing purposes.
- **Biology:** It can be used for classification among different species of plants and animals.
- **Libraries:** It is used in clustering different books on the basis of topics and information.
- **Insurance:** It is used to acknowledge the customers, their policies and identify the frauds.
- **City Planning:** It is used to make groups of houses and to study their values based on their geographical locations and other factors present.

- **Earthquake studies:** By learning the earthquake-affected areas we can determine the dangerous zones.
- **Search Engines:** The search result appears based on the closest object to the search query.

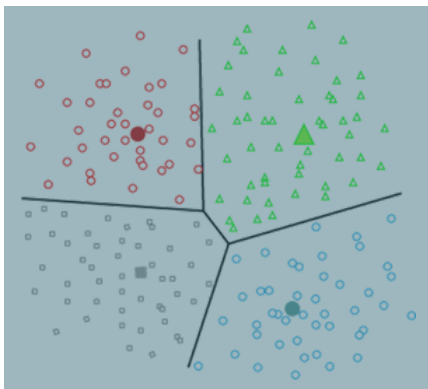
Types of Clustering Methods

The clustering methods are broadly divided into Hard clustering (datapoint belongs to only one group) and Soft Clustering (data points can belong to another group also). But there are also other various approaches of Clustering exist. Below are the main clustering methods used in Machine learning:

- **Partitioning Clustering**
It is a type of clustering that divides the data into non-hierarchical groups. It is also known as the centroid-based method.

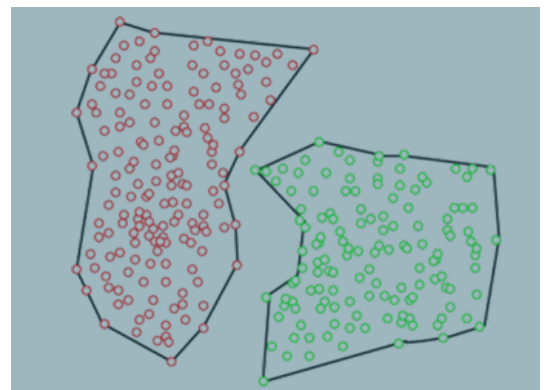
This is basically one of the iterative clustering algorithms in which the clusters are formed by the closeness of data points to the centroid of clusters. Here, centroid is formed such that the distance of data points is minimum with the center and the number of clusters to be created is specified.

Algorithm example is K-Means



- **Density-Based Clustering**
The density-based clustering method connects the highly-dense areas into clusters, and the arbitrarily shaped distributions are formed as long as the dense region can be connected. This algorithm does it by identifying different clusters in the dataset and connects the areas of high densities into clusters. The dense areas in data space are divided from each other by sparser areas.

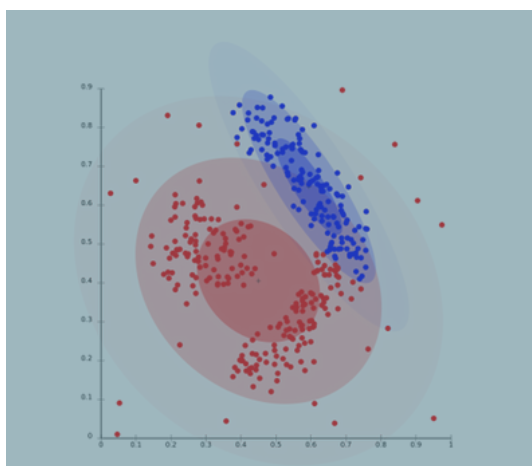
Distribution-based clustering produces clusters that assume concisely defined mathematical models underlying the data, a rather strong assumption for some data distributions. Algorithm example is Gaussian Mixture Model (GMM)



- **Distribution Model-Based Clustering**

It is a clustering model in which we will fit the data on the probability that it may belong to the particular distribution. The grouping is done by assuming some distributions, commonly Gaussian Distribution.

Distribution-based clustering produces clusters that assume concisely defined mathematical models underlying the data, a rather strong assumption for some data distributions. Algorithm example is Gaussian Mixture Model (GMM).



- **Fuzzy Clustering**

It belongs to a branch of soft method clustering techniques, whereas all the above-mentioned clustering techniques belong to hard method clustering techniques. In this clustering technique, points close to the center may be a part of the other cluster to a higher degree than points at the same cluster's edge. The probability of a point belonging to a given cluster is a value that lies between 0 to 1.

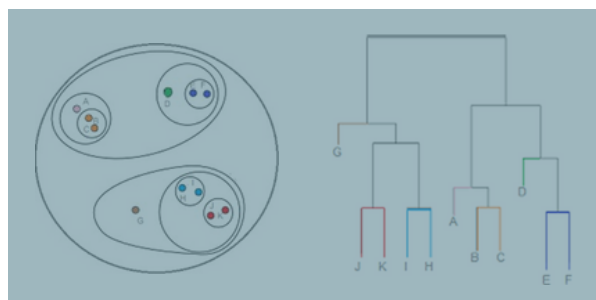
Algorithm examples: Fuzzy C-Means



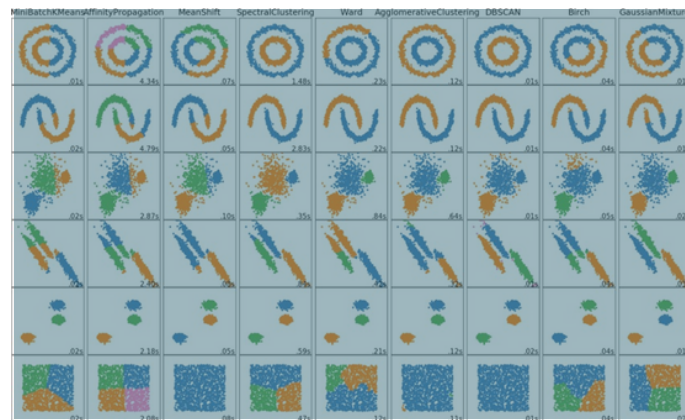
- **Hierarchical Clustering**

It can be used as an alternative for the partitioned clustering as there is no requirement of specifying number of clusters. It is also known as connectivity-based methods. In this method, it provides us with the hierarchy of the clusters that merge after a certain distance. After the clustering is done, the result will be a tree-based representation of data points (Dendrogram), divided into clusters.

Number of clusters can be selected by cutting the tree at the correct level. Algorithm examples are BIRCH, Affinity propagation, Agglomerative.



Different Algorithms on Different Kinds of Data

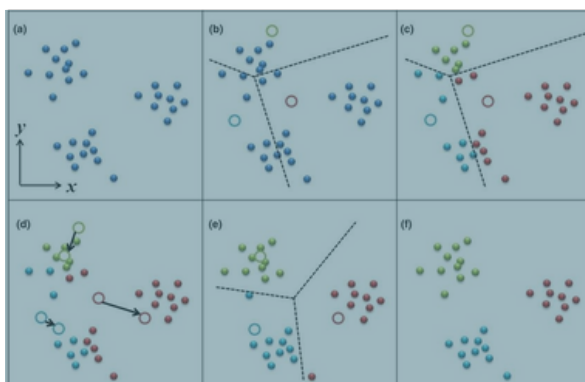


Please note that you don't have to remember all the algorithms, just know that there are multiple algorithms with different use cases, so you can just check them later and then compare to see which one is more suitable for your problem?

K-Means Algorithm

The goal of this algorithm is to find groups in the data, with the number of groups represented by the variable K . The algorithm works iteratively to assign each data point to one of K groups based on the features that are provided. Data points are clustered based on feature similarity. The end results of the K-means clustering algorithm would be:

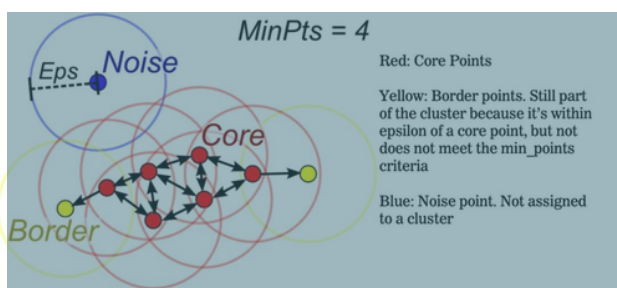
- Centroids of the number of clusters, which were identified (denoted as K).
- Labels for the training data.



DBSCAN Algorithm

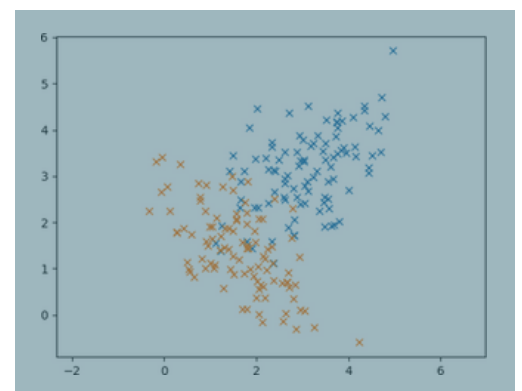
It groups 'densely grouped' data points into a single cluster. It can identify clusters in large spatial datasets by looking at the local density of the data points. The most exciting feature of DBSCAN clustering is that it is robust to outliers. It also does not require the number of clusters to be told beforehand, unlike K-Means, where we have to specify the number of centroids.

DBSCAN requires only two parameters: epsilon and minPoints. Epsilon is the radius of the circle to be created around each data point to check the density and minPoints is the minimum number of data points required inside that circle for that data point to be classified as a Core point.



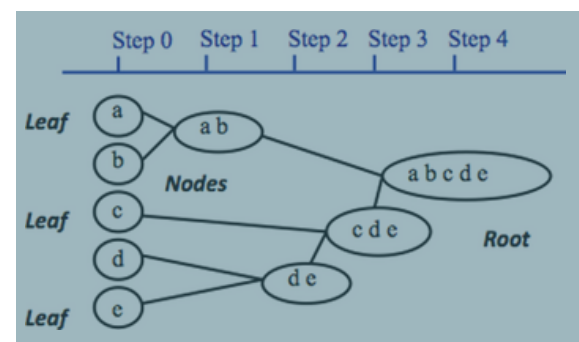
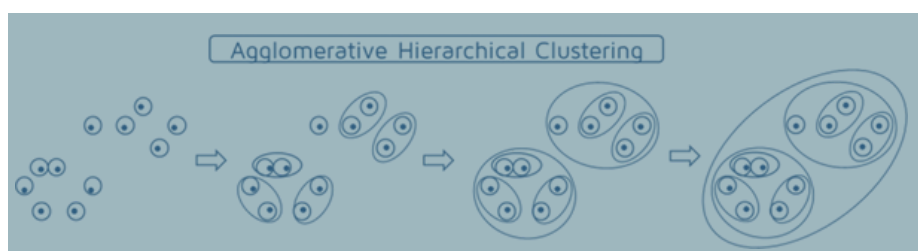
Gaussian Mixture Algorithm

Consider there is a case where we have overlap in our data and K-Means doesn't seem to work well. Suppose we know the distribution and it is Gaussian, we can use this info and use Gaussian Mixture algorithm.



Agglomerative Algorithm

Agglomerative clustering works in a "bottom-up" manner. That is, each object is initially considered as a single-element cluster (leaf). At each step of the algorithm, the two clusters that are the most similar are combined into a new bigger cluster (nodes). This procedure is iterated until all points are members of just one single big cluster (root). The steps can be:



Picture Source

- [pexels.com](https://www.pexels.com)